

Yi Shi

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EXPERIENCE

Software Development Engineer | Cactus Capital Group LLC, TX / Remote

Sep. 2025 – Present

CapEx – Investment Portfolio Analytics Platform

- Architected a full-stack portfolio analytics platform using **React.js**, **TypeScript**, and **Express.js** with **AWS DynamoDB** and **Redis** caching, delivering real-time P&L dashboards to institutional clients at sub-100ms query latency.
- Engineered **Python** and **Node.js** microservices on **AWS Lambda** for automated market data ingestion, risk assessment, and portfolio rebalancing alerts, processing millions of daily transactions with 99.9% uptime.
- Built **JWT**-authenticated **RESTful APIs** with **MySQL** and role-based access control; integrated **GitHub Actions CI/CD** pipelines, cutting deployment cycle time by 35% across enterprise client environments.

Software Development Engineer Intern | TDK SensEI, PA

Jun. 2025 – Aug. 2025

Bigswing – GenAI Anomaly Prediction Service

- Designed a production-grade **GenAI** anomaly prediction service integrating **ML inference** via **AWS Bedrock**, **Lambda**, and **S3**, processing high-frequency sensor data with 99%+ precision to prevent manufacturing failures.
- Built real-time monitoring dashboards with **React.js**, **Express.js**, and **RESTful APIs**; refactored inference pipeline from Bedrock to **SmolAgent / LangChain**, reducing response latency by 30%.
- Built synthetic data and load-testing frameworks, cutting end-to-end pipeline runtime by 95% and validating system stability under edge cases.

Software Development Engineer Intern | Shawk Solution, WA

Jan. 2025 – Apr. 2025

CowBot – Gemini-based AI Querying Agent

- Designed a production **Gemini**-based AI querying agent with **Node.js / Express.js** APIs enabling natural-language data access; integrated **Pinecone** vector database for embedding retrieval, reducing query latency by 30%.
- Built a scalable monorepo frontend with **React**, **TypeScript**, and **Tailwind CSS** via **Nx**; migrated backend to **Bun** for 40% throughput gain; secured APIs with **JWT / GraphQL**-based role-based access control.
- Collaborated with UI/UX teams via **Figma** and **Miro** to design and deliver responsive, accessible interfaces.

Software Development Engineer | Joblogic-X Corporation, TX

Jun. 2021 – Feb. 2023

CarbonX – Carbon Trading & Data Platform

- Optimized carbon trading data pipelines on **AWS** with **S3**, **EMR**, **Airflow**, and **Parquet**, achieving 12x faster processing using **Spark DataFrames** and **Spark SQL** on large-scale datasets.
- Containerized services with **Docker** and deployed fault-tolerant microservices via **Kubernetes** RollingUpdate; upgraded **CI/CD** pipelines with **MongoDB** logging and **Elasticsearch** for zero-downtime releases.

MetricsDash – Real-Time Analytics Dashboard

- Built interactive analytics dashboards with **Python**, **Flask**, and **Dash**; integrated **Elasticsearch** for full-text search in high-concurrency environments, enhancing data retrieval.
- Implemented **Pylint**, **Pytest**, **Selenium**, and **Jest** test suites ensuring API and UI reliability across 20+ endpoints.
- Integrated **MongoDB** data sources for structured log querying and real-time chart updates on the dashboard.

Land Management System – ArcGIS Platform

- Engineered a real-time **ArcGIS** land management system with **Vue.js**, **TypeScript**, and **Vuex** (20% LCP/INP/CLS improvement); secured via **Java / Spring Boot** APIs and **Redis-over-MySQL** caching (15% faster).
- Built interactive map features with **ArcGIS JS API** and **TypeScript** for real-time property visualization and client management.
- Spearheaded a VR exhibition app with **Unity**, **C#**, **Unity XR Toolkit**, **Oculus**, and **Pico SDK** for immersive cross-platform 3D experiences for 1000+ users.

EDUCATION

Northeastern University – Master of Science in Computer Science

Boston, MA | Dec. 2025

Denison University – Bachelor of Science in Biology; Computer Science Minor; Art Studio Minor

Columbus, OH | Jun. 2021

SKILLS

Expert: Python, Java, TypeScript, JavaScript, React.js, Node.js, Express.js, Angular.js, MySQL, MongoDB, Redis, Amazon DynamoDB, AWS, Claude Code, GitHub

Proficient: Spring Boot, Spring Cloud, PostgreSQL, SQL Server, Flask, Django, Next.js, Vue.js, MyBatis, Docker, Kubernetes

Familiar: C, C#, C++, Go, React Native, Ruby, Shell, Scala, Spark, Hadoop, Apache Airflow, SVN